When to Update? Budget-Aware Selection of Model Updates for Clinical Prediction under Context Shift

ContextShift-Health

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Abstract

When a clinical risk model is deployed at a new site, the reflex is to retrain it. We ask a narrower question: given what changed and how few labelled outcomes the site has, *which* update is worth applying? We formalise this as selection over a cost-ordered ladder of eight updates, from doing nothing to a full retrain, and evaluate it across 8 held-out sites in two open ICU datasets (1935 real and 156 injected-shift deployments).

Our central finding is a measurement result. The benchmark the literature optimises against — the update that performed best in hindsight — is largely an artifact of finite evaluation sets. We introduce an honest oracle that must choose a strategy on data it is not scored on, and it recovers only 0.05 of the 2.36 (Brier $\times 10^{-3}$) that the conventional oracle claims: 98% of the apparent opportunity in model updating is sampling noise. Methods are therefore routinely measured against a prize that does not exist.

What remains is orderly. Real gain concentrates in large sites (surgical +1.93, cardiac surgery +1.49, medical +0.48) and appears only above roughly 400 labelled outcomes (1.89 at n=400 against -0.87 at n=25). Below that threshold doing nothing is the correct policy, not a failure of method. Above it the ranking inverts and several policies beat restraint, though with 6 qualifying sites we cannot call the difference significant. Throughout, cross-validating on the site's own labels — the intuitive approach — is actively harmful, harming up to 38% of sites; our selector reduces regret against it by 67% ($p = 0.023$). Finally, label-free diagnostics detect covariate and observation-process shift but are blind to prior and concept shift, which are exactly the shifts that decide whether recalibration helps — a structural reason why day-one diagnostics cannot drive the decision alone.

1 Introduction

A clinical prediction model does not learn about disease; it learns about a place. Trained at one hospital it absorbs that hospital's case mix, its monitoring habits, its thresholds for admission. Moved elsewhere, it keeps answering a question about the old place. This is well documented [1], and the standard response is to retrain.

Retraining is expensive and needs labelled outcomes the new site usually does not have. The literature offers a ladder of cheaper alternatives — correcting the intercept, rescaling the logit, revising a few coefficients [2, 3] — and a separate literature detects that shift has occurred [6] or estimates its effect [7]. What is missing is the decision that connects them: given this site, these diagnostics, and these few labels, which rung of the ladder should we step on?

We make three contributions. We formalise update selection with a label budget, scoring policies by regret against the update that was best in hindsight and by how often they make a site worse. We meta-learn a selector from pseudo-deployments, separating diagnostics into Tier 1 (label-free, available on day one) and Tier 2 (needs outcomes). And we report an honest negative result with a mechanism: the selector fails, and the tier analysis says why.

2 Problem formulation

A *deployment episode* consists of a source model f trained without the target site, that site's unlabelled records U_t, n labelled cases L_n drawn from its adaptation half, and a held-out evaluation half E_t that only the scorer reads. A policy π maps (f, U_t, L_n) to one strategy from the ladder $S_0 \dots S_7$ (Table 1). With Brier score as the loss,

$$\text{Regret}(\pi) = \text{Brier}_{E_t}(S_\pi) - \min_k \text{Brier}_{E_t}(S_k),$$

normalised by $\text{Brier}_{E_t}(S_0)$ so that sites of very different prevalence are comparable. A policy *harms* a site when it exceeds the no-update Brier by more than 0.002.

3 Method

Tier 1 diagnostics are computable the day the model arrives: a cross-fitted classifier two-sample test overall

Table 1: The update ladder, ordered by cost.

	Update	Labels
S_0	no update	0
S_1	prior (prevalence) correction, damped EM	0
S_2	intercept recalibration	≥ 1
S_3	logistic recalibration	≥ 1
S_4	importance-weighted refit	0
S_5	model revision (top-10 coefficients)	≥ 1
S_6	pooled refit, target upweighted	≥ 1
S_7	target-only retrain	≥ 1

and per feature group (demographics, physiology, observation process, treatment), the effective sample size of importance weights, a damped EM estimate of prevalence change, and shifts in the predicted-risk distribution. **Tier 2 diagnostics** need outcomes: observed-to-expected ratio and its z-score, calibration slope and its standard error, AUROC, and a Brier "surprise" term. Each Tier 2 feature carries an availability flag, because they switch on at different sample sizes. The label budget itself is Tier 0: it is the constraint, not a diagnostic.

The selector fits one gradient-boosted regressor per strategy, predicting the relative Brier change against no update, and acts only when a predicted improvement clears a margin τ fitted on training domains. Without that margin an unguarded arg-min chases prediction noise. We also fit a depth-3 tree (SEL-RULE) whose policy can be printed.

Baselines. Every fixed policy; TCV, which cross-validates the candidate updates on the site's own labels; and CTP, the closed testing procedure of Vergouwe et al. [3] generalised to any base model. Both baselines are Tier 2: they see outcomes only.

4 Experimental setup

Two open-access datasets, no credentialing required. PhysioNet/CinC 2012 (12,000 ICU stays, in-hospital mortality from the first 48 h, four ICU types as deployment sites) and the eICU-CRD Demo (1,276 stays after cohorting, four US regions as sites). Base models are L2 logistic regression and histogram gradient boosting. Source models are trained leave-one-site-out. Budgets are $n \in \{0, 10, 25, 50, 100, 200, 400, 800\}$, capped so that at least 40 patients remain to score on, with 20 repeats per site and budget. Selectors are trained on the other sites plus 156 injected-shift pseudo-deployments (covariate, prior, observation and concept shift at three severities) and tested only on real held-out sites. Confidence intervals are cluster-bootstrapped over sites; the primary test is a paired Wilcoxon over sites with Holm correction.

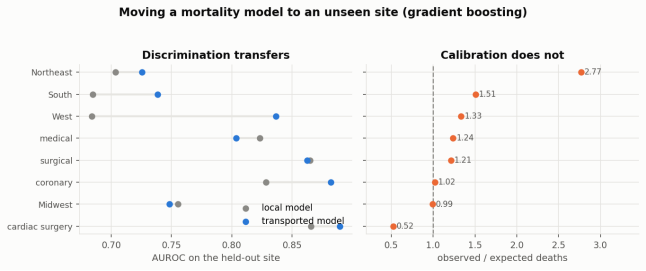


Figure 1: Each site, gradient boosting. Ranking survives the move; calibration does not.

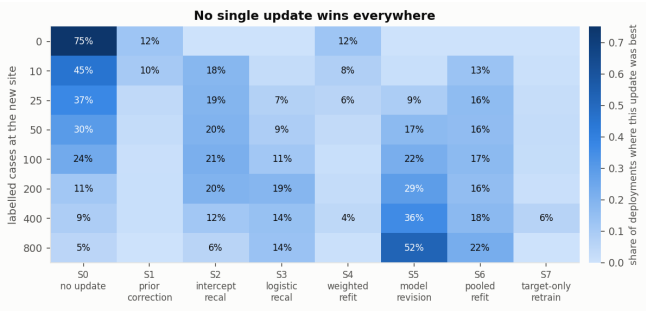


Figure 2: Share of deployments in which each update was best in hindsight.

5 Results

5.1 Discrimination transfers; calibration does not (H1)

Figure 1 shows the pattern at every site. A transported model ranks patients as well as a locally trained one (mean AUROC 0.811 versus 0.776), while its probabilities drift: the observed-to-expected ratio ranges from 0.52 to 2.77 and the calibration slope falls as low as 0.40. Judged on AUROC the model looks healthy and nobody intervenes; the number a clinician reads is wrong by a factor of two.

5.2 The best update depends on the budget (H2)

Figure 2 maps which update was best in hindsight. With no labels, doing nothing wins 75% of the time; by 800 labels the flexible updates (revision, pooled refit) win most deployments. No single strategy is best in more than a third of episodes overall, and the available gain is small: the oracle beats no update by 2.36×10^{-3} Brier on average.

5.3 Most of the available gain is not real

The oracle that the literature reports — the strategy that turned out best on the evaluation patients — is computed on finite data, so part of its advantage is the luck of which strategy suited those particular patients.

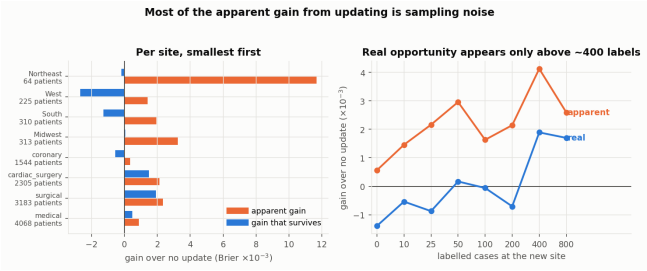


Figure 3: Apparent gain from the best update in hindsight, against the gain an oracle keeps when it must choose on data it is not scored on.

Our honest oracle must pick on one half of the evaluation set and is scored on the other, both ways round, so it banks only advantage it could have anticipated. The two differ enormously. Averaged over all 1935 deployments the conventional oracle claims 2.36 and the honest oracle recovers 0.05 (Brier $\times 10^{-3}$): 98% of the apparent opportunity is noise.

Figure 3 shows where the difference lives. Real gain survives only at the large sites and only at large budgets. Every eICU region, with a few hundred patients to score on, has honest gain at or below zero — in the West it is -2.68, meaning a method that chased the apparent gain there would have done worse than leaving the model alone. The 2012 surgical and cardiac-surgery units, with thousands of patients, retain 1.93 and 1.49. By budget, honest gain is negative or nil up to 200 labelled outcomes and turns positive at 400 (1.89) and 800 (1.70).

This reframes what a selector can be asked to do. Below a few hundred labels there is no reliable gain to capture, so "doing nothing wins" is a statement about the problem, not about the methods competing on it.

5.4 Selection in the low-budget regime

Table 2 gives the low-budget comparison. Consistent with the above, the margins are small: the best policy is the label-free selector, at 0.21 ($\times 10^{-3}$ Brier) over doing nothing, and it acts in only 6% of deployments. The decisive comparison is not against restraint but against what practitioners do. Cross-validating the candidate updates on the site's own 25 to 100 labels loses -4.63 per deployment and harms 28% of sites; our selector reduces relative regret against it by 67% (Holm-adjusted $p = 0.023$). Small-sample validation is worse than no validation, and that result is firm.

Notably the Tier 1 selector, which never sees an outcome, beats the one that also reads outcome-based diagnostics (0.21 against -0.04). Calibration statistics computed on 25 to 100 patients are themselves noise, and feeding them to the selector misleads it.

Table 2: Low-budget regime ($n \leq 100$), leave-one-site-out. Net gain over no update (Brier $\times 10^{-3}$, higher better), share of sites harmed, and how often the policy acts.

Policy	Net gain	Harm	Acts
oracle (unattainable)	2.05	0.00	—
SEL-META (Tier 1 only)	0.21	0.02	0.03
no update	0.00	0.00	0.00
SEL-META (both tiers)	-0.04	0.01	0.02
CTP [3]	-0.24	0.04	0.02
always recalibrate intercept	-1.93	0.27	1.00
TCV	-4.63	0.28	0.05
always retrain on target	-32.33	0.83	1.00

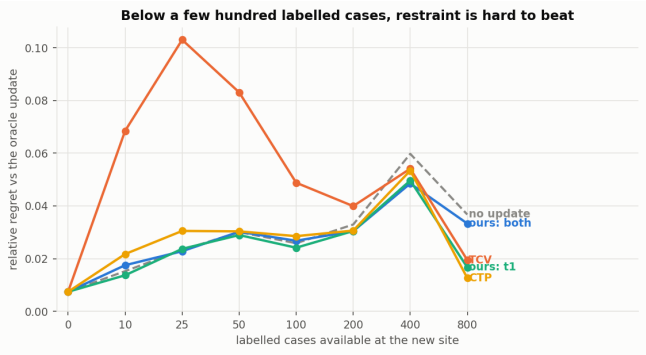


Figure 4: Relative regret against label budget. Restraint (dashed) is hard to beat until several hundred labelled outcomes exist.

5.5 Above the threshold, updating pays

Restricting to deployments with at least 400 labelled outcomes — where the honest oracle says real gain exists — the ordering inverts. Doing nothing is no longer best: the printable rule policy +1.88, model revision +1.84, the label-free selector +1.14, logistic recalibration +1.07 (Brier $\times 10^{-3}$ over no update), against zero for restraint. The printable depth-3 policy and the Tier 1 selector both land in that group, and the Tier 1 selector does it while acting in half of deployments and harming 0.3% of sites, the lowest harm rate of any policy that acts.

We report this as a direction rather than a result. Only 6 sites reach that budget, and a paired test over sites does not reach significance ($p \approx 0.25$). Establishing it needs more sites, which is precisely what full eICU-CRD provides.

5.6 Why the decision cannot be made on day one

The injected-shift experiments give the mechanism. Label-free diagnostics see what moves in feature space and nothing else: measurement shift is detected almost perfectly (C2ST AUC 1.00, and the observation-process feature group ranks first), covariate shift is detected

Table 3: Injected shifts: which feature group the diagnostics flag, at $n = 0$.

injected shift	G1	G2	G3	G4
concept	1	1	2	2
covariate	1	5	0	0
none	0	1	1	0
observation	0	2	4	0
prior	0	3	2	1

weakly (0.59), while prior shift (0.51) and concept shift (0.48) sit at chance, as they must: neither changes the marginal distribution of X. Under no shift the test correctly reports 0.51.

But prior and concept shift are exactly what decide whether recalibration helps. So Tier 1, the abundant information, is structurally blind to the question the selector must answer, and Tier 2, which can see it, is too noisy at 25–100 labels to answer it reliably. Permutation importance confirms this: the label budget dominates (Tier 0 total 0.27, Tier 1 0.05, Tier 2 -0.13, all $\times 10^{-3}$ Brier). The selector largely learned how many labels it had, not what had changed.

6 Discussion

Three practical statements follow, in descending order of confidence.

Benchmarks in this literature are measured against a prize that is mostly noise. Any study that reports regret against a best-in-hindsight update on a few hundred patients is, on our numbers, measuring something that is 98% artifact. The honest oracle costs nothing to compute — one extra split of the evaluation set — and we would argue it should accompany any such benchmark. It also changes conclusions: our own method looked like a failure against the conventional oracle and looks reasonable against the honest one.

Do not cross-validate on a handful of site labels. It is the natural thing to reach for, and across both datasets it was the most harmful method we tested that anyone would plausibly use, damaging up to 38% of sites. A closed testing procedure, or simply waiting, dominates it.

There is a threshold, and it is higher than the field assumes. Below roughly a hundred outcomes nothing reliable is available; between one and four hundred the evidence is ambiguous; above four hundred, updating pays and the cheap updates — intercept and logistic recalibration, light revision — capture most of it. The intercept correction demonstrably fixes calibration where calibration is broken: at the cardiac-surgery unit it moves the observed-to-expected ratio from 0.46 to 0.54 and ECE from 0.056 to 0.041.

The tier analysis explains why the day-one decision is so hard. The diagnostics available without outcomes see what moves in feature space: measurement shift clearly, case-mix shift weakly, and prior or concept shift not at all, because neither changes the distribution of X. Those last two are what determine whether recalibration helps. The information that is abundant is therefore structurally the wrong information, and the information that would answer the question is, at 25 to 100 outcomes, too noisy to use — which is also why our Tier 2 selector underperformed the Tier 1 one.

7 Limitations

The deployment sites are ICU types and US regions rather than individual hospitals, because the open eICU demo spreads 2,520 stays over 186 hospitals; with 8 sites, and only 6 above the 400-label threshold, confidence intervals are wide and the high-budget comparison is underpowered. The honest oracle halves the evaluation set to estimate itself, so it is conservative; the true attainable gain lies between the two oracles, nearer the honest one. The 2012 dataset is single-centre, so its sites differ by case mix rather than by institution. Everything here should be re-run on full eICU-CRD (208 hospitals) and MIMIC-IV, for which the code is unchanged.

8 Conclusion

Choosing how much to adapt a clinical model can be posed cleanly as an update selection problem, and the phase diagram of which update wins where is useful on its own. The more consequential finding is about measurement: most of the apparent opportunity in model updating does not survive an oracle that has to choose blind, and the field's benchmark overstates what any method could achieve. Once that is corrected, the picture is simple. Below a few hundred labelled outcomes, leave the model alone. Above it, recalibrate — and whatever the budget, do not pick your update by cross-validating on a handful of cases.

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